MAIS PAC

IPhO 03.02: Uncertainty Analysis · Worked Solutions

# Uncertainty Analysis: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IPhO questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

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**Solution 1 — Mean and standard deviation.** Return →

(a) $x = (12.4+12.6+12.5+12.3+12.7)/5 = 12.50$ {mm}.

(b) Deviations: -0.1, +0.1, 0, -0.2, +0.2. Sum of squares = 0.01+0.01+0+0.04+0.04 = 0.10, so

$$s = \sqrt{(0.10/4)} = 0.16 \text{ {mm}}. \quad (1)$$

(c) $\sigma_x = 0.16/\sqrt{(5)} = 0.07$ {mm}, so the result is (12.50 ± 0.07) {mm}. **Check:** the readings scatter by about ±0.2 {mm}, and the mean is more reliable than any single reading — exactly what 1/√()N predicts.

**Solution 2 — Simple propagation.** Return →

(a) $P = 2(a+b) = 20.0$ {cm}; the uncertainties add, $2(0.1+0.2) = 0.6$ {cm}, or $2\sqrt{(0.1^2+0.2^2)} = 0.45$ {cm} in quadrature: (20.0 ± 0.5) {cm}.

(b) $A = ab = 24.0$ {cm²}. Fractional: 0.1/4.0 = 2.5% and 0.2/6.0 = 3.3%, so $\Delta A/A = \sqrt{(2.5^2+3.3^2)} = 4.2\%$, giving (24.0 ± 1.0) {cm²}.

(c) The area multiplies two quantities, so their relative uncertainties add; the perimeter adds lengths, so the **absolute** uncertainties add, but the perimeter is five times larger, making its relative uncertainty small. **Check:** 4.1% of 24 is about 1 {cm²}, consistent with the worst-case value 0.1×6 + 0.2×4 = 1.4 {cm²} being a little larger.

**Solution 3 — A density measurement.** Return →

(a) $V = \pi d^2 h/4 = \pi(2.00)^2(3.00)/4 = 9.42$ {cm³}, so $\rho = 84.2/9.42 = 8.94$ {g,cm^(-3)}.

(b) Fractional uncertainties: mass 0.1/84.2 = 0.12%, diameter 0.1/20.0 = 0.50% (doubled to 1.0%), length 0.2/30.0 = 0.67%. Total $\sqrt{(0.12^2+1.0^2+0.67^2)} = 1.2\%$, so $\rho = (8.94 \pm 0.11)$ {g,cm^(-3)}.

(c) The diameter and the length contribute about equally, and the mass hardly at all. Improving the diameter measurement (a micrometer instead of a vernier) would cut its contribution to 0.1%, leaving the length as the limit. **Check:** the value is close to copper (8.96 {g,cm^(-3)}), which is a useful sanity check.

**Solution 4 — Timing and the pendulum.** Return →

(a) $T = 100.4/50 = 2.008$ {s}, with uncertainty $0.2/50 = 0.004$ {s}: $(2.008 \pm 0.004)$ {s}.

(b) $g = 4\pi^2 L/T^2 = 4\pi^2(1.000)/(2.008)^2 = 9.79$ {m,s^(-2)}. Fractional: $2(0.004/2.008) = 0.20\%$ from T and 0.2% from L, so $\Delta g/g = 0.28\%$ and $g = (9.79 \pm 0.03)$ {m,s^(-2)}.

(c) The accepted value 9.81 {m,s^(-2)} differs by 0.02, well inside the 0.03 uncertainty — the results agree. **Check:** the measured value is slightly low, and in a real experiment this is often because the pendulum is a physical pendulum (extended bob) or because the amplitude was not small.

**Solution 5 — Gradients from a graph.** Return →

(a) $g = 4\pi^2/m = 4\pi^2/4.05 = 9.74$ {m,s^(-2)}. The uncertainty in the gradient is (4.11-3.99)/2 = 0.06, so $\Delta g/g = 0.06/4.05 = 1.5\%$ and $g = (9.74 \pm 0.15)$ {m,s^(-2)}.

(b) The best-fit line gives an intercept of +0.02 {s²} on the $T^2$ axis.

(c) Theory says the intercept should be zero, so a positive intercept of 0.02 {s²} is about 1% of the $T^2$ values — probably a small systematic effect, such as a reaction-time delay in starting and stopping the watch, or an error in measuring L from the wrong point. **Check:** an intercept in $T^2$ corresponds to a constant delay, which is exactly what a stopwatch reaction time produces.

**Solution 6 — Combining independent uncertainties.** Return →

(a) f = 10.0×20.0×5.0 = 1000. Fractional: 5%, 1%, 10%; combined √(25+1+100) = 11.2%, so f = (1000 ± 110).

(b) h = 10.0+20.0-5.0 = 25.0 with Δ h = √(0.5²+0.2²+0.5²) = 0.73, so h = (25.0 ± 0.7) — a relative uncertainty of 2.9%.

(c) In h the absolute uncertainties add but the value is a sum of the magnitudes, so the **relative** uncertainty is diluted; in f the relative uncertainties add, which is much more punishing. **Check:** 110/1000 = 11% against 0.7/25 = 2.9%, as stated.

**Solution 7 — Worst case versus quadrature.** Return →

(a) Worst case: 2% + 2% = 4%.

(b) Quadrature: √(2²+2²) = 2.8%.

(c) Worst case assumes both errors conspire in the same direction, which is unlikely if they are independent random errors; quadrature is the expected value for independent random contributions. Examiners usually accept either, provided the reasoning is stated, but the quadrature result is the standard for random errors, and worst case is used for tolerances that could genuinely add (manufacturing limits, for instance). **Check:** quoting a smaller uncertainty than justified is a serious error; quoting a larger one is merely conservative.

**Solution 8 — The lens experiment.** Return →

(a) The intercept equals 1/f, so f = 1/0.0500 = 20.0 {cm}, with Δ f/f = 0.0010/0.0500 = 2%: f = (20.0 ± 0.4) {cm}.

(b) The gradient is -0.98 ± 0.03, so -1 lies within about 0.7 of the uncertainty — consistent with the theory, though slightly shallow.

(c) A gradient shallower than -1 suggests that the measured object and image distances are not exactly the distances from the lens: distances measured from the lens holder's centre, or a lens of non-negligible thickness, both produce such an offset. **Check:** the intercept method is more robust than measuring f directly, because it uses all the data points at once.

**Solution 9 — A resistivity measurement.** Return →

(a) $A = \pi(0.25\times10^{-3})^2 = 1.96\times10^{-7}\ \{m^2\}$, so

$$\rho = (RA)/(L) = ((2.14)(1.96\times10^{-7}))/(1.000) = 4.2\times10^{-7}\ \Omega,\{m\}. \qquad (2)$$

(b) Fractional: 0.02/2.14 = 0.9%, 2(0.01/0.50) = 4.0%, 0.005/1.000 = 0.5%; combined 4.1%, so $\rho = (4.2 \pm 0.2)\times10^{-7}\ \Omega,\{m\}$.

(c) The diameter alone contributes 4.0% of the 4.1%: it dominates completely, because it is squared and because 0.01 {mm} on 0.50 {mm} is a 2% reading error. **Check:** the value is close to nichrome (about $1.1\times10^{-6}$) or to a carbon-steel alloy; the exact identity matters less than the honest uncertainty.

**Solution 10 — Deciding whether results agree.** Return →

(a) Difference = 0.4 Ω; combined uncertainty = $\sqrt{(0.1^2+0.2^2)} = 0.22\ \Omega$.

(b) The discrepancy is 0.4/0.22 = 1.8 times the combined uncertainty, which is under 2 — the results are consistent within errors, though not comfortably so.

(c) A third, more precise measurement would shrink the uncertainties and show whose value (if either) is off; if the two original measurements are both slightly mis-calibrated in opposite directions, the average of the two would be right by accident. "They agree within errors" must be justified by the calculation above, not asserted, because a claim of agreement is a quantitative statement about the overlap of two ranges. **Check:** had the difference been 0.8 Ω, the ratio would be 3.6 and a systematic error would be the natural explanation.